MANOJ DHAMALA

Santa Clara, CA | (350) 930-1032 | manojktm9999@gmail.com | www.linkedin.com/in/manojktm9999/

EDUCATION

University of the Cumberlands **Williamsburg, KY**
MS in Data Science Expected May 2026

Salem State University **Salem, MA**
Bachelors in IT May 2022

SKILLS

ML & AI: LLMs (LoRA fine-tuning), NLP, Deep Learning, VAE, Dimensionality Reduction, Predictive Modeling (Regression, Decision trees), Classification, Clustering, Feature Engineering, Model Evaluation
Languages & Tools: Python, SQL, PyTorch, scikit-learn, HuggingFace, Pandas, NumPy, Git, Linux, CUDA, MLflow, Airflow, GCP, AWS, Power BI

EXPERIENCE

Trine LLC **Bellevue, WA**
Junior Data Scientist Nov 2023 – Feb 2025

- Led a Customer Demand Forecasting project using regression and decision tree models in Python and scikit-learn, improving weekly forecast accuracy by ~18% and supporting inventory planning decisions.
- Built a Customer Segmentation model using clustering techniques on 120K+ records, helping marketing refine targeting strategy and increase campaign response rates by 12%.
- Fine-tuned a transformer-based LLM using LoRA (PyTorch, Hugging Face) for internal document summarization, reducing manual review time and lowering inference costs by 30%.
- Developed reusable feature engineering pipelines with Pandas and NumPy and applied PCA for dimensionality reduction, cutting model training time by 25% across multiple projects.
- Implemented automated model tracking and scheduled retraining workflows using MLflow and Airflow, improving experiment reproducibility and deployment reliability in AWS and GCP environment.
- Created executive-facing Power BI dashboards backed by SQL data extraction, reducing reporting turnaround from several days to same-day delivery and improving cross-team visibility.

Ampex Brands **Richardson, TX**
Data Analyst May 2022 – Oct 2023

- Led an Inventory Optimization Project across multiple franchise locations by querying sales and supply data with SQL and analyzing trends in Python (Pandas, NumPy). Built regression models in scikit-learn to forecast weekly ingredient demand, reducing stockouts and lowering excess inventory by 14%.
- Managed a Seasonal Promotions Performance Analysis for limited-time menu items across KFC and Panera Bread locations. Used classification models to predict promotion success based on pricing, region, and past campaign data, helping operations adjust rollout strategies and improve promotional ROI by 11%.
- Built a Store-Level Sales Segmentation Model using clustering techniques to group locations by sales patterns and customer traffic. Shared findings through Power BI dashboards, helping regional managers adjust staffing and ordering plans by store type.
- Developed a Menu Mix Profitability Model using decision trees and feature engineering to identify which product combinations drove higher margins during peak hours. Results informed cross-selling strategies that increased average ticket size across select stores.
- Created an automated Daily Reporting Pipeline using Python, Airflow, and MLflow to track KPIs such as food cost percentage, waste levels, and promo lift. Reduced manual reporting time and improved consistency across brand leadership reports.
- Supported a Customer Feedback NLP Project by analyzing survey and review text using NLP techniques with Hugging Face and PyTorch. Applied dimensionality reduction to summarize themes and helped operations address recurring service and product issues across franchises.

PROJECTS

Customer Demand Forecasting Platform Nov 2023 – Apr 2024

- Built regression and decision tree models using Python and scikit-learn to forecast weekly demand across product lines, improving forecast accuracy by ~18% and reducing planning gaps.
- Engineered time-based and behavioral features with Pandas and NumPy, and tracked experiments using MLflow to compare model performance.
- Deployed the forecasting workflow in AWS, working with engineering to support scheduled retraining using Airflow.

LLM-Powered Internal Document Summarization April 2024 – Jan 2025

- Fine-tuned a transformer model using LoRA (PyTorch, Hugging Face) to summarize long internal reports, cutting analyst review time by 30%.
- Applied dimensionality reduction to evaluate embedding quality and improve summary consistency across departments.
- Version-controlled training scripts with Git and ran experiments in a Linux/CUDA environment to support efficient model training.